

Mini Project

Business Problem Statement

Scenario

You have recently joined ^{PAK}~~Amazon India~~ as a **Junior Data Analyst**.

The Sales and Operations team has provided you with sales transaction data from different states, product categories, payment modes, and customers.

The management wants to understand the overall business performance before planning the next quarter's strategy.

As a Data Analyst, your responsibility is to:

- Clean the dataset.
- Analyze sales and profit.
- Identify top-performing products and categories.
- Find underperforming regions and products.
- Create dashboards and visualizations.
- Present business recommendations.

Your analysis should help management answer key business questions and make data-driven decisions.

PART 1 – Excel Tasks

Dashboard

Students should include

Profit is not be calculated as the Dataset not have any relevant column

$\text{Profit} = \text{Cost} - \text{Revenue}$

(but Cost is not given)

KPI Cards

- Total Sales
- ~~Total Profit~~

- Total Orders
- Average Sales
- ~~Average Profit~~



Charts

- Bar Chart
- Sales by State
- Column Chart
- Profit by Category
- Pie Chart
- Sales by Category
- Line Chart
- Monthly Sales Trend



Slicers

- State
- Category
- Payment Mode



Business Questions (Students Must Answer)

1.

What is the **total sales**, ~~total profit~~, and **average order value**?

2.

Which **state** generated the highest and the lowest sales?

3.

Which **category** and ~~sub-category~~ contributed the highest revenue?

4.

Which are the **Top 10 products** based on sales?

5.

Who are the **Top 5 customers** contributing the highest revenue?

6.

Which **payment mode** is most preferred by customers?

7.

Which **month** recorded the highest sales and profit? Is there any noticeable sales trend?

 8.

Identify any ~~categories or products with high sales but low profit.~~ What could be the possible reason?

9.

Which **state or city** requires improvement in sales performance? Suggest possible business strategies.

10.

Based on your complete analysis, provide **three actionable business recommendations** that Amazon's management can implement to improve sales and profitability.



Dataset Loading

Q1. Load the Amazon_Sales.xlsx dataset into a Pandas DataFrame and display the first five records.

✓ Dataset Overview

Q2. Display:

- Number of rows and columns
 - Column names
 - Data types
 - Statistical summary of numerical columns
-

✓ KPI Analysis

Q3. Calculate the following KPIs:

- Total Sales
 - ~~Total Profit~~
 - Total Orders
 - Average Sales
 - ~~Average Profit~~
 - Maximum Sales
 - Minimum Sales
-

✓ Sales Analysis

Q4. Find the total sales generated by each **State** and display the results in descending order.

Q5. Find the total sales generated by each **Category**.

Q7. Find the total sales generated by each **Sub-Category**.

✓ Customer & Product Analysis

Q9. Display the **Top 5 Customers** based on total sales.

Q10. Find the total quantity sold for each product.

✓ Payment Analysis

Q11. Find the number of orders placed using each **Payment Mode**.

✓ Time-based Analysis

Q12. Calculate the monthly sales trend using the **Order Date** column.

✓ Multi-level Analysis

Q13. Find the sales generated by each **State** for every **Category**.

(Hint: Use `groupby()` with multiple columns.)

✓ Visualization

Q15. Create a **Bar Chart** showing Sales by State.

Q16. Create a **Pie Chart** showing Category-wise Sales.

Q17. Create a **Line Chart** showing Monthly Sales Trend.

Q18. Create a **Horizontal Bar Chart** showing the Top 10 Products by Sales.